

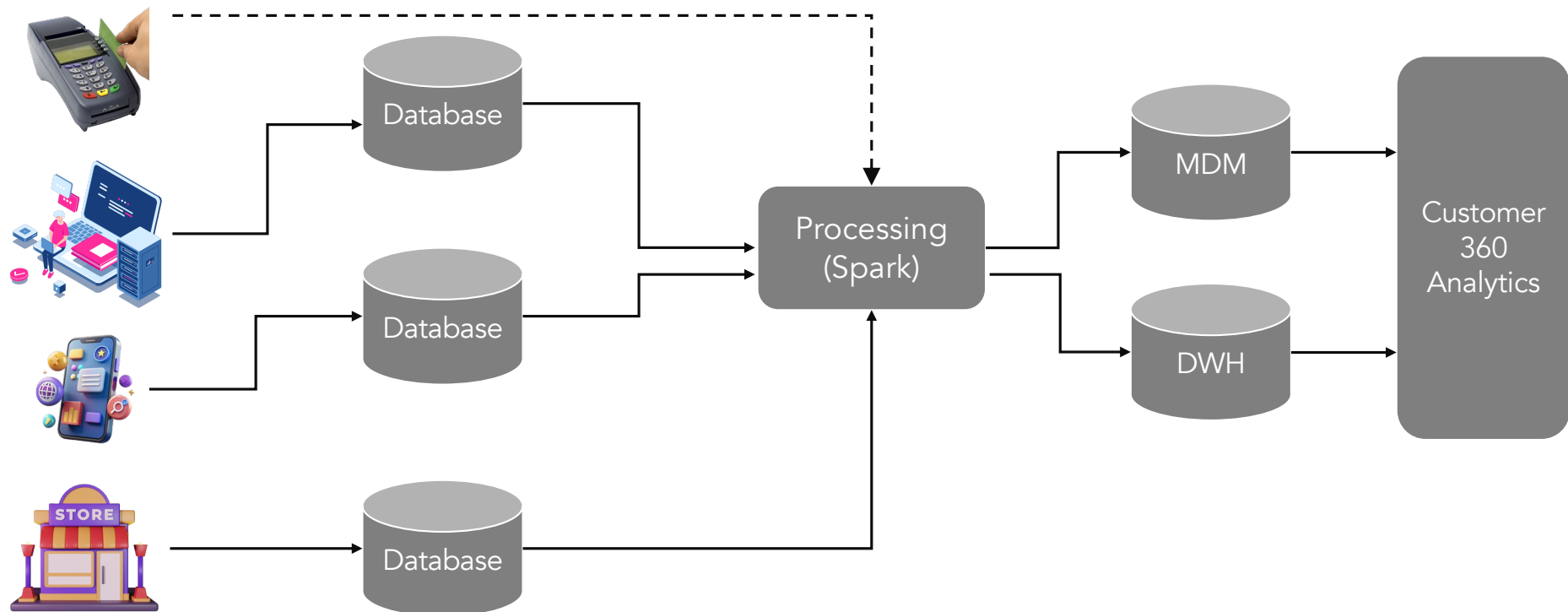
Customer 360 Application

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Customer 360 Analytics

- Applicable to all type of organizations.
- Customers interact with organization from various communication points – Mobile App, Web Application, POS machines, Retail Store etc.
- The purpose of Customer 360 Analytics is to collect these data points from different communication points or sources and put them in a single “source of truth”.
- The final goal is to create a view of consumer behavior like location, interaction across communication points, purchase history.
- Purpose:
 - Targeted personalized marketing to create new opportunities.
 - Predictive analytics and decision support systems (DSS).
 - Improve customer experience and retention.
 - Increase cross-selling, upselling, product development.

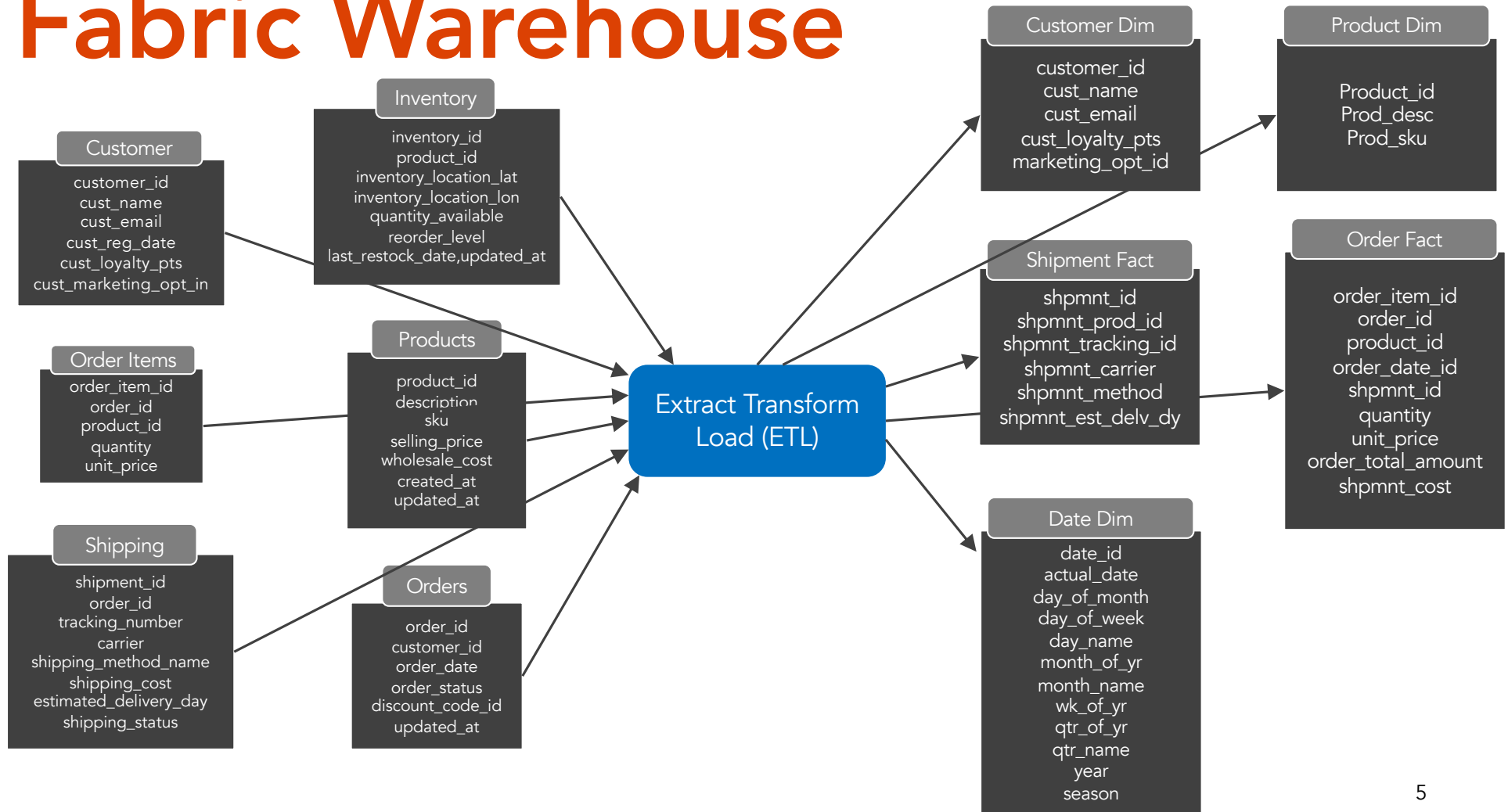
Customer 360 Tech Stack



Source Databases (MySQL)

- CSV Files:
 - Customer Data
 - Inventory Data
 - Order Items Data
 - Orders Data
 - Product Data
 - Shipping Data
- Uncompressed data size – 35 GB
- Task 1 – Load the files into individual tables in MySQL/PostgreSQL.

Fabric Warehouse



Fact Table Details

Column Name	Description
shpmnt_id	Shipment Id
shpmnt_prod_id	Product of the shipment. A single shipment should contain all the products for an order.
shpmnt_tracking_id	Shipment tracking number
shpmnt_carrier	Shipment carrier
shpmnt_method	Shipment Method
shpmnt_est_delv_dy	Estimated delivery day

Column Name	Description
order_item_id	Order item id
order_id	Order id
product_id	Product id
order_date_id	Order date
shpmnt_id	Shipment id
quantity	Quantity of each product
unit_price	Price of individual product
order_total_amount	Cost of order containing all products
shpmnt_cost	Shipment cost

- Task 2 – Load fact and dimension tables from the source tables

Dimension Table Details

Column Name	Description
date_id	Unique Id of a row
actual_date	Actual date in YYYY-MM-DD.
day_of_month	Day of a particular month of actual date. 1 to 31.
day_name	Name of day_of_month. Monday to Sunday.
month_of_yr	Month of the year in actual date. 1 to 12.
month_name	Name of the month. January to December.
wk_of_yr	Week of year that actual date falls in. 1 to 52.
qtr_of_yr	Quarter of actual date. 1 to 4.
qtr_name	Name of quarter. Q1, Q2, Q3 & Q4
year	Year of actual date.
season	Season of actual date. Summer, Winter, Autumn, Spring.

- Task 3 – Populate Date Dimension table as per the column description. Start and End date should be 1 year below and above the min and max order date respectively. Use Python or PySpark DATE functions.

Business Statements – Task 4.1

- Use Data Warehouse to solve these business queries.
- Find out the distribution of sales of top 15 products every year.
- Which shipping carrier is the slowest each year. How shipping carriers have improved delivery lead time over the years. Delivery lead time = estimated_delivery_day.
- What is the avg. delivery time of each method for each carrier. Which carrier performs best in Express delivery and which carrier performs best in Standard delivery.
- What is the distribution of sales of products on weekdays vs weekends.
- Find out top 10 frequent buyers and their avg. transaction value.
- Find out top 10 frequent buyers and their avg. transaction value for marketing opt in and opt out separately. Do you see any trend of the products that they buy.
- What is the total lifetime buy value for top 50 customers for each marketing opt in and opt out category.

Business Statements – Task 4.2

- Fix loyalty points for each customer based on total purchase value. For a purchase amount of 100, 2 loyalty points should be awarded. Fix this in Customer Dimension table.
- Get monthly sale from each inventory for each individual year.
- Find total order count for each quarter of each year. Which season has the highest sale each year.